Deep anisotropic $LiNbO_3$ etching with SF_6/Ar inductively coupled plasmas

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A SF_6/Ar inductively coupled plasma (ICP) technique was investigated to improve etching of proton exchanged $LiNbO_3$. The influences of He backside cooling, power, and gas flows on characteristics such as etching rate, sidewall slope angle, and surface roughness were investigated. Total gas flow is a key parameter that affects etching results, and an optimized gas flow (50 sccm) was used for lengthy etching processes (30 min). Deep ($>3\ \mu$m) and highly anisotropic etching, as well as ultra smooth $LiNbO_3$ surfaces were achieved in a single-step run. The authors' proposed method has achieved the deepest, most vertical, minimal residue structure yet reported for single-step ICP etching. © *2012 American Vacuum Society*. [DOI: 10.1116/1.3674282]

I. INTRODUCTION

Lithium niobate ($LiNbO_3$) is a widely used dielectric material, and is extremely important in integrated and nonlinear optical devices because of its large electro-optic coefficients,[1] large transparency range, and wide intrinsic bandwidth. It can be used to fabricate many integrated optical devices, such as electro-optic modulators, optical amplifiers, and integrated laser gain/conversion components.[2] Waveguides in $LiNbO_3$ crystals are traditionally fabricated by metal diffusion, annealed proton exchange processes,[3] and ion irradiation.[4] These methods, while well-established, have the drawbacks of small index contrast yields and subsequently, weak guiding. Large turning radii are needed for such waveguides and many $LiNbO_3$ devices are consequently much larger than they would otherwise need to be. Dry etching was investigated here as an alternative method to overcome these limitations on the utility of $LiNbO_3$.

Deep waveguides can provide better optical confinement than channel waveguides. Deep ridge waveguides have low loss and can be bent sharply (because of greater index contrast) to reduce component size. Consequently, deep, vertical etching is a key strategy for realizing future compact integrated devices. The etching profile of a waveguide must be considered and carefully controlled during deep etching processes. Vertical sidewalls and rectangular structures allow fabrication of waveguides with better mode profiles and reduced insertion loss, which are especially important in multimode interference (MMI) and Mach–Zehnder interferometers (MZI).[5] The ability to realize low-loss rectangular waveguide structures would lead to improved performance of MZI modulators, by enabling the use of photonic crystals, ring resonators, etc., which require a high index contrast not achievable with traditional $LiNbO_3$ waveguide fabrication processes. In this report, we focus on $LiNbO_3$ etching, but it is also worth pointing out that high index contrast vertical confinement is also often desirable for low-loss optical waveguiding, and that monolithic techniques do exist to create slabs of $LiNbO_3$.[6] Although in this study we etched bulk $LiNbO_3$, the ICP methods under investigation could also apply for such slabs.

Wet etching and dry etching are the two techniques most commonly used to etch $LiNbO_3$ substrates. Wet etching, based on a mixture of HF and HNO_3, is a simple process for etching $LiNbO_3$. Wet etching, combined with a variety of methods, including proton exchange,[7] ion implantation,[8,9] metal deposition,[10,11] and domain inversion[12] has been reported to fabricate ridge $LiNbO_3$ waveguides in recent decades. However, wet etching is a time consuming process due to its very low etching rate. The poor anisotropic property of wet etching is also a drawback. Dry etching has many advantages over wet etching because it is highly controllable and has highly anisotropic properties. Thus, dry etching is widely used in microfabrication applications. Several dry etching techniques have been reported for etching $LiNbO_3$, such as direct laser writing,[13] FIB milling,[14] neutral loop discharge plasma (NLD),[15] and reactive ion etching (RIE).[16,17]

Recently, ICP etching has drawn increased attention for $LiNbO_3$ because of the greater control possible over the resultant etch profile relative to other methods. An ICP system has two RF units and can control plasma density and ion energy independently, which means it can be operated at low chamber pressure and still achieve a high etching rate and high anisotropy. ICP etching with different gases including CHF_3/Ar, CF_4, SF_6/Ar and CF_4/He has been studied previously to improve the etching rate in $LiNbO_3$.[18–21] In this study, we also use the proton exchange (PE) technique to reduce the concentration of Li ions in the $LiNbO_3$ surface,

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leading to less redeposition of LiF and an improved etching rate. Because $LiNbO_3$ is a difficult material to etch, there have been no prior reports of deeply-etched anisotropic structures (>2 μm) realized in a single etch step. L. Gui *et al*.[22] recently reported a relatively deep etch (3 μm depth) using ICP with a C_4F_8 and He gas mixture. Their process was lengthy (>40 min) and samples needed to be taken out and cleaned in SC-1 (RCA) solution to remove a brown layer. Additionally, the $LiNbO_3$ substrate was very rough after etching and another annealing process was required to smooth the surface. The main reason why it is hard to realize deep structures in $LiNbO_3$ is the formation of LiF during the etching process.[23] LiF has a high boiling point of 1676 °C. In addition, metal particles sputtered by ion bombardment and other fluoride-based inhibitors that get deposited on the $LiNbO_3$ surface during etching serve to prevent further etching after a given time. For deep structure etching, the redeposition of such inhibitors is a critical issue to overcome. As etching time increases, inhibitors will accumulate on the surface and sidewalls causing the etching rate to decrease significantly. Consequently, careful control over the etching conditions is very important for lengthy etching processes. In addition, the deposition of a thick metal mask is also a critical issue. The mask needs to be thick enough to prevent strong physical sputtering, which is necessary to remove the residue.

In this work, we investigated the details of the ICP etching process on PE-$LiNbO_3$ samples under different etching conditions and realized deep structures on bulk $LiNbO_3$ in a single etching step. SF_6/Ar gas mixtures were used. Standard photolithography and electron beam evaporation were first applied to create a metal etching mask on the sample surface. After etching, the surface condition, etching rate and profile were characterized by a surface profiler and scanning electron microscopy (SEM). The effects of ICP conditions on the resulting surface roughness and sidewall slope angles were investigated, in order to better obtain deep and cleanly-etched structures. In addition, we demonstrated the dependences of etching rate, surface condition and anisotropy on different ICP parameters.

II. EXPERIMENTAL CONDITIONS

Etching studies were performed on Z-cut $LiNbO_3$ wafers cut into $1.4 \times 1\,cm^2$ samples. $LiNbO_3$ samples were first immersed in molten benzoic acid at 235 °C for 5 h to perform proton exchange (PE). During the PE process, hydrogen ions will in-diffuse into samples and change the $LiNbO_3$ surface's composition into $H_xLi_{1-x}NbO_3$, down to a certain depth. After standard cleaning with acetone, isopropanol and deionized water, $LiNbO_3$ samples were spin-coated using a photoresist (AZ5214E), and photolithography was carried out to pattern various stripes of 2 – 18 μm width on the sample surface. Cr was used as the etching mask due to its high hardness (Mohs hardness of Cr is 8).[24] This helps prevent strong ion bombardment during etching and shows high selectivity. In addition, in our previous work[25] we showed that Cr can lead to a better surface condition (post-etch) and also can be easily removed compared to other metals using commercial Cr etchant. The etch selectivity of proton exchanged $LiNbO_3$ to the Cr mask was 12:1, which is suitable for this deep $LiNbO_3$ etching study. Here, a 300 nm Cr film was deposited on the $LiNbO_3$ samples using an Edwards Auto 306 electron beam evaporator and patterned through a lift-off process. Unfortunately, the condition of our photoligthography mask and lift-off process did not yield as smooth a metal stripe side as desired, and this will later be visible as sidewall roughness.

Plasma etching was conducted in a PlasmaTherm SLR 770 ICP system. An 8-inch Si carrier wafer was used to move the samples from the loadlock into the reactor chamber. Before etching, O_2 plasma was applied for 15 min to clean the chamber. The gas composition affects the ultimate etching rate, surface smoothness, and anisotropy significantly. Ar was added to the fluorine-based gas to increase the physical etching component and enhance the anisotropy. As reported in Ref. 18, SF_6 can lead to a smooth surface condition and reduced inhibitor formation. Consequently, we chose SF_6/Ar as the gas mixture for these etching studies.

During a preliminary study of short etching processes, we found that the etching profile can be optimized at 7.0 mTorr chamber pressure, 90 W RIE power and 800 W ICP power. A perfect rectangular structure with a sidewall slope angle of 90° was achieved, as shown in Fig. 1. The etching time was 20 min, and samples were etched in two steps (10 min each) to obtain 1.2 μm total depth. Note that on both sides of the ridge, etched micro-trenching can be observed. This micro-trenching is due to ions glancing off sidewalls of the ridge structure resulting in additional etching at the edges. For later long-duration single-step etching studies, we always kept the ICP power consistent at 800 W and chamber pressure at 7.0 mTorr, and other etching conditions were varied. After ICP etching, etching rates were obtained by measuring the depth of etched patterns with a surface profiler.

III. RESULTS AND DISCUSSION

A. Effect of He backside cooling

The effects of He backside cooling and two different etching modes (intermittent and continuous) were studied.

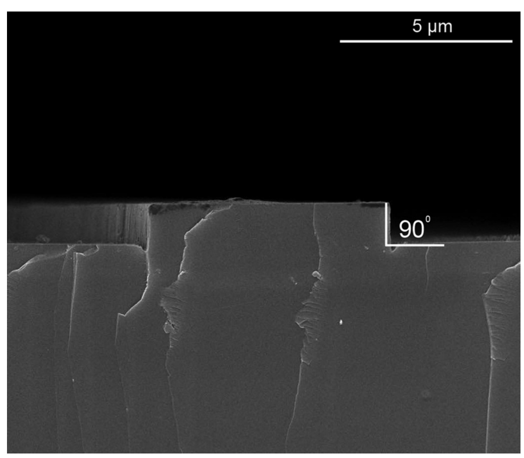

FIG. 1. SEM cross-sectional image of a perfect rectangular etched structure with 90° slope angle at 90/800 W (RIE/ICP power), 7 mTorr pressure.

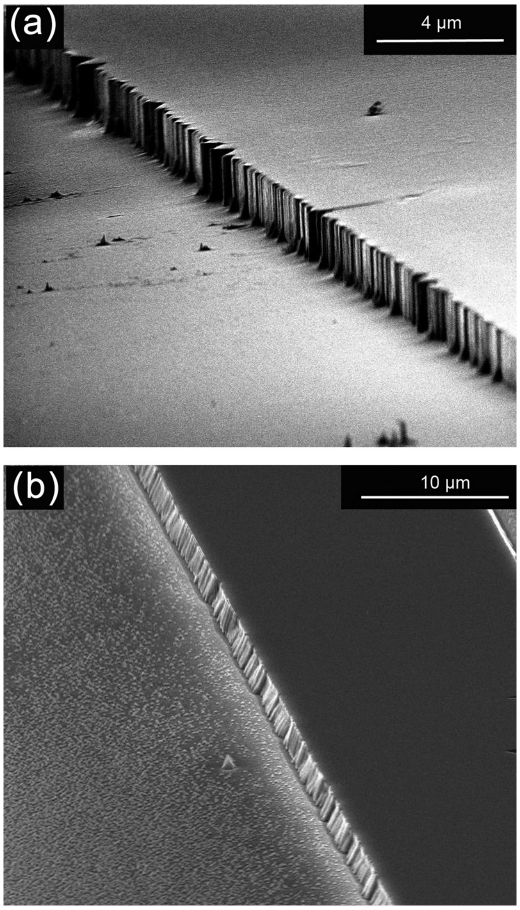

FIG. 2. SEM images of etched $LiNbO_3$ surface condition with (a) continuous mode, and (b) intermittent mode.

The same ICP parameters were used for all samples: the ratio of SF_6 to Ar was fixed at 1:1, the ICP power was 800 W, RIE power was 90 W, chamber pressure was 7.0 mTorr and total gas flow was 10 sccm. For intermittent etching, there was a 2 min break after every 10 min of etching. During the etch break, only Ar was introduced into the reactor chamber to maintain the chamber pressure, and all RF power was off. The total etching time was fixed at 30 min. For continuous etching, the etching process was performed continuously for 30 min. Continuous mode etching leads to a higher wafer surface temperature than intermittent mode, in which etching breaks allow some cooling of the wafer to occur.

A He cooling system integrated into the wafer chuck is commonly used for wafer backside heat dissipation. Turning off the He cooling system also will lead to an increase in wafer temperature. Etching results show that etching rate can be significantly improved by turning off the He backside cooling system and using continuous mode. The etching rate for the intermittent process increases from 47 nm/min with He cooling to 55.2 nm/min without He cooling, and the etching rate for the continuous etch process increases from 50 nm/min with He cooling to 61 nm/min without He cooling. For $LiNbO_3$ etching, the main byproducts are NbFx, especially NbF_5. Either etching without He cooling or etching under continuous mode will cause the substrate temperature to increase significantly and help in the evaporation of NbF_5 and also assist in conversion of NbF_4 into NbF_5. Consequently, the etching rate will be increased significantly.

Figure 2 shows an SEM image of the surface condition (after etching) for intermittent and continuous modes. Intermittent mode etching resulted in a rough surface, as shown in Fig. 2(b). In contrast, continuous etching for 30 min [Fig. 2(a)] produced a smooth surface. The sidewall slope (nearly vertical) after continuous etching was also much more favorable than after intermittent etching.

B. Effect of RIE power

Figure 3 shows the measured etching rate as a function of RIE power. All other etching parameters were kept constant.

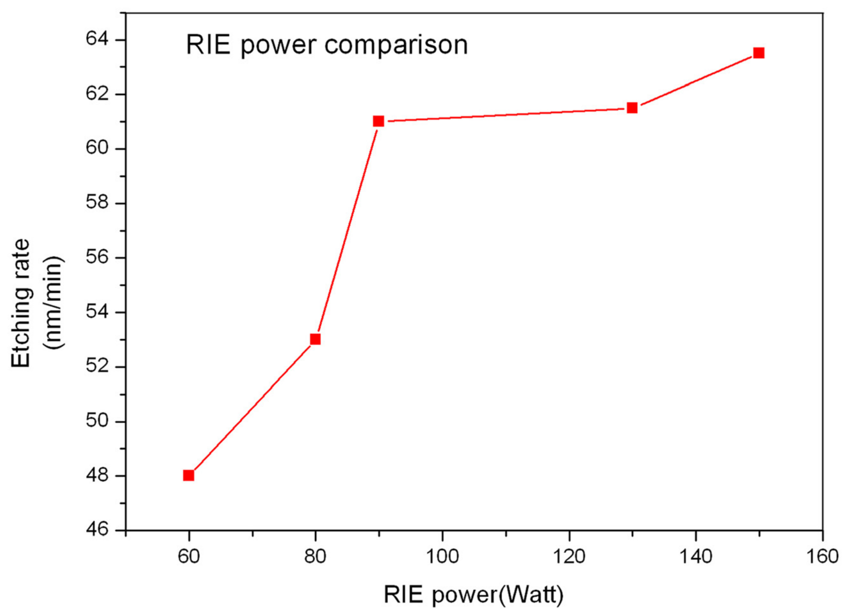

FIG. 3. (Color online) $LiNbO_3$ etching rate as a function of RIE power.

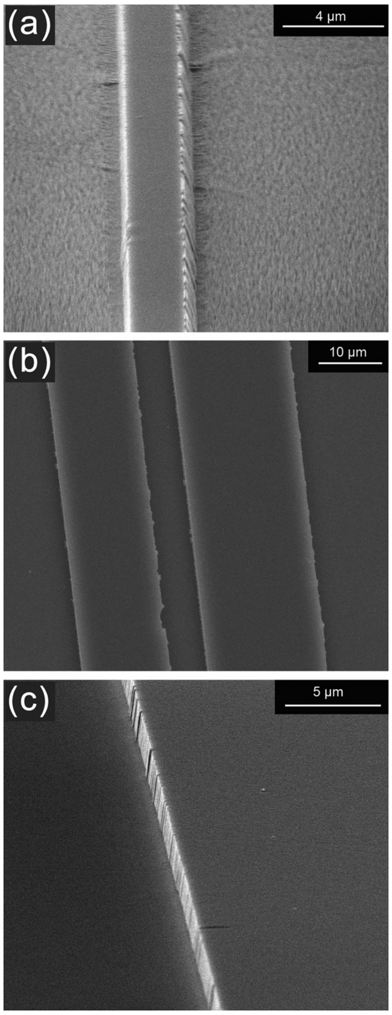

FIG. 4. SEM images of etched $LiNbO_3$ surface condition with different RIE powers: (a) 60 W, (b) 90 W, and (c) 150 W.

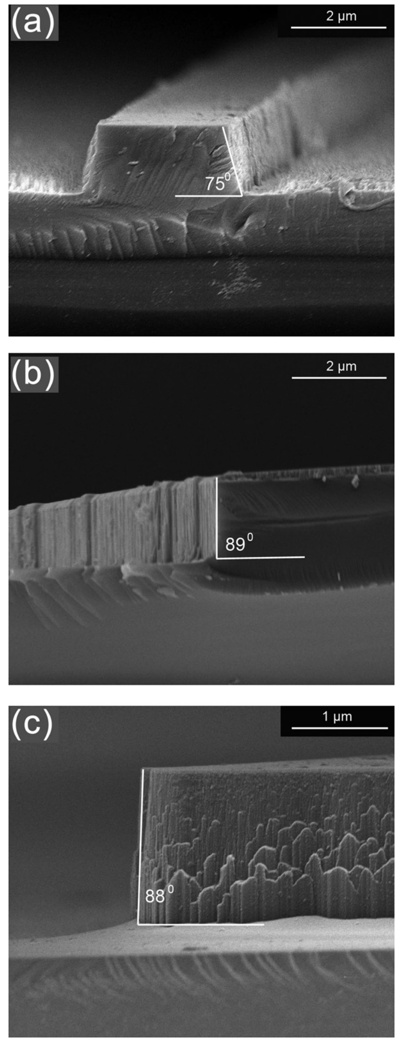

FIG. 5. SEM images of etched $LiNbO_3$ cross section with RIE power at (a) 60 W, (b) 90 W, and (c) 150 W. Slope angles are shown.

The etching time is 30 min in continuous mode. RIE power varied from 60 W to 150 W. It can be seen that etching rate increases slightly as RIE power increases. An etching rate of 63.5 nm/min was achieved at 150 W RIE power. This can be explained by the fact that higher RIE power will cause higher DC bias voltage and higher ion energy, corresponding to strong ion bombardment and more efficient bond breaking in the $LiNbO_3$ substrate; thus increasing the etching rate.

Figure 4 shows an SEM image of the $LiNbO_3$ surface condition after etching with different RIE powers. It can be seen that the surface conditions were less favorable at lower RIE power. This is due to insufficient ion energy (lower RIE power corresponds to lower DC bias voltage). When the RIE power is too low, weak ion bombardment cannot sputter away all inhibitors and etching residues on the surface, which then serve as micromasks, leading to a rougher surface.[26]

Increasing the RIE power can also influence etching anisotropy, as shown in Fig. 5. At 60 W RIE power, an etched slope angle of 75° was obtained [Fig. 5(a)], while powers of 90 W and 150 W resulted in etched slope angles of 89° and 88°, respectively [Figs. 5(b) and 5(c)].

C. Effect of total gas flow and deep etching

Figure 6 presents the influence of total gas flow on the etch rate of $LiNbO_3$. The ratio of SF_6 to Ar was fixed at 1:1. It can be seen that the etching rate of $LiNbO_3$ is affected strongly by the total gas flow. There is a minimal difference in the etching rate between 10 sccm and 30 sccm. However, when gas flow was increased to 50 sccm, the etching rate improved significantly. The etching rate of 98.6 nm/min at 50 sccm gas flow was achieved for SF_6/Ar gas, which is almost double that achieved at 30 sccm gas flow, 48 nm/min. It was difficult to maintain the chamber pressure at 7.0 mTorr when a total gas flow of 100 sccm was used. During the etching process, the actual chamber pressure increased to 8.5 mTorr. It seems the etching rate decreased slightly when 50/50 sccm (SF_6/Ar) gas flow was used.

Figure 7 shows the cross section of etched $LiNbO_3$ strips at different total gas flow rates. It can be seen that the total gas flow influences the etching anisotropy significantly. The etch structure with a highest slope angle (~90°) was obtained at a total gas flow of 10 sccm [Fig. 7(a)]. As the

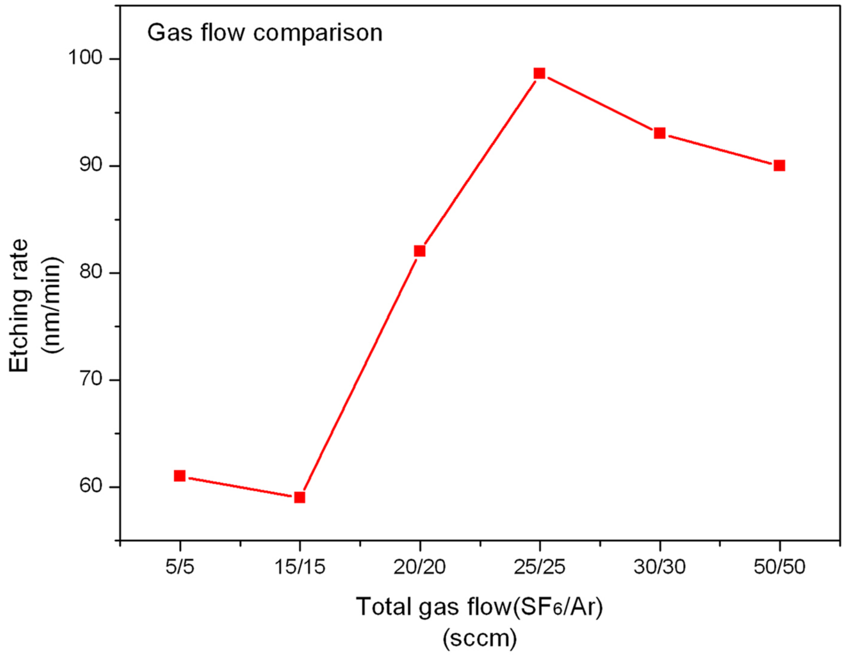

FIG. 6. (Color online) $LiNbO_3$ etching rate as a function of total gas flow. The ratio of SF_6 and Ar is fixed at 1:1.

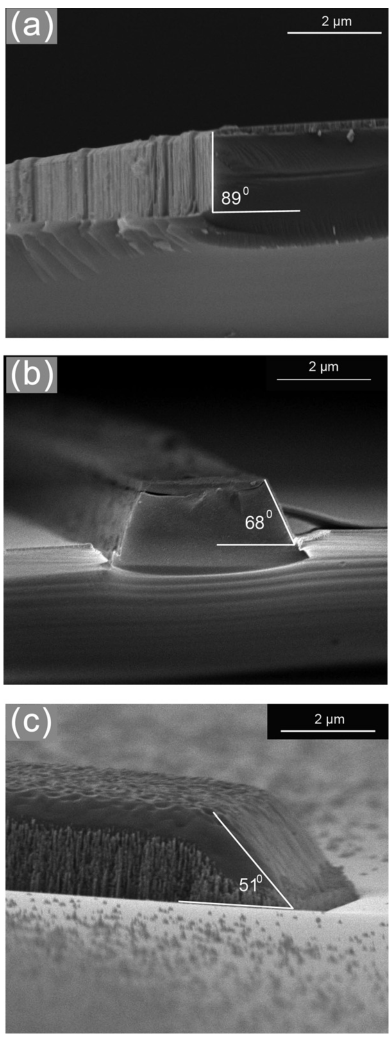

FIG. 7. SEM cross-sectional images of etched $LiNbO_3$ with slope angle at total gas flows at (a) 10 sccm, (b) 30 sccm, and (c) 60 sccm.

total gas flow increased, the slope angle decreased to 51° [Fig. 7(c)]. The inhibitor mechanism can explain the influence of gas flow on etching anisotropy.[27] During etching, the fluorine reacts with $LiNbO_3$ and forms a nonvolatile inhibitor like LiF. With an increase of the gas flow, the formation of this inhibitor is enhanced. During lengthy etching processes, more and more inhibitor is deposited on the sidewall, impeding further etching there, while the inhibitor deposited on the bottom is removed by ion bombardment. Consequently, the sidewall slope angle decreases as gas flow increases. This phenomenon is much more significant in long-duration etching. In addition, notice that the surface of the $LiNbO_3$ substrate becomes rough at 30/30 sccm SF_6/Ar gas flow [Fig. 7(c)], while a smooth surface was obtained at lower gas flow [Figs. 7(a) and 7(b)]. The rough surface is also mainly due to the enhancement of the inhibitor. With an increase of the gas flow, the redeposition of the inhibitor (especially LiF) is enhanced greatly. The speed of redeposition of LiF is faster than the speed of physical sputtering. This indicates that the ion bombardment cannot remove all residues.

With relatively high gas flow and optimized etching conditions, structures with deep plasma LN etching (3.5 μm), ultra-smooth surfaces and high anisotropy were achieved, as shown in Fig. 8. This is the deepest and most vertical low-residue $LiNbO_3$ structure yet demonstrated using ICP in a single-step etching. As is clear in Fig. 8, trenching phenomenon appeared on both sides of the ridge. This was due to additional etching by ion reflection and charging of the sidewall.[28] Figure 8 shows the nearly vertical but very rough sidewalls which were mainly caused by imperfect photolithography mask and lift-off processes. Ladroue *et al.* showed that sidewall roughness can be reduce by using a SiO_2 mask instead of a metal mask,[29] which will be considered for future studies.

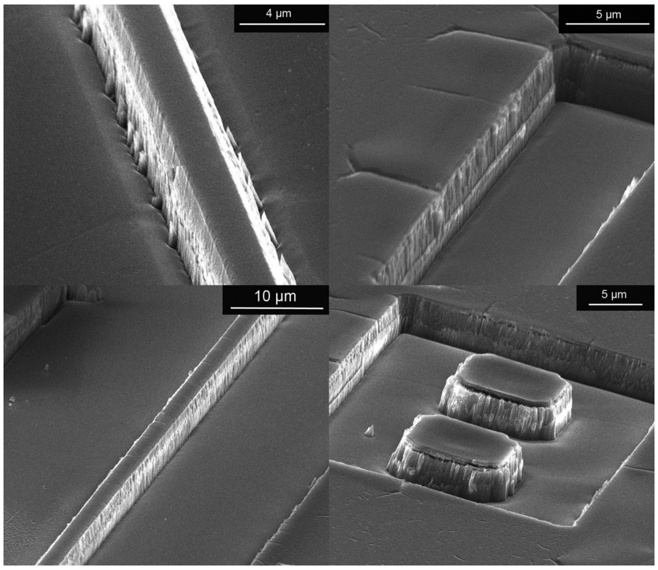

FIG. 8. SEM images of deep and highly anisotropic $LiNbO_3$ etching at 3.5 µm depth under 800 W ICP power, 90 W RIE power, 7 mTorr pressure, and 50 sccm SF_6/Ar gas mixture. The etching time is 30 min.

IV. CONCLUSIONS

Proton exchanged $LiNbO_3$ etching was studied in SF_6/Ar plasmas as a function of process parameters. The effects of He backside cooling, RIE power and total gas flow were investigated. The results show that higher etching rate and a smooth post-etch surface condition can be obtained without He cooling. Higher RIE power can improve the etching rate and surface condition, also leading to highly anisotropic etching. The total gas flow can significantly affect the etching rate, anisotropy and surface roughness. The etching rate can be improved to 98.6 nm/min when the total gas flow increases to 25/25 sccm (SF_6/Ar), while the slope angle will decrease from 89° at 5/5 sccm SF_6/Ar to 51° at 30/30 sccm SF_6/Ar. Deep ($>3\,\mu m$) and highly anisotropic PE $LiNbO_3$ structures with ultra-smooth surfaces were achieved in a single-step run, with parameters of 30 min at 800 W ICP power, 90 W RIE power, 7 mTorr chamber pressure and 50 sccm total gas flow. The results show promise that deep etching on $LiNbO_3$ may be an attractive method to realize a variety of structures and fabricate optical and MOEMS devices.

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